

Deduce before you label: strategy attribution for large language model agents in the iterated prisoner’s dilemma

Author One¹ and Author Two^{1,2}

¹Affiliation One

²Affiliation Two

Abstract

Language models are routinely said to play tit-for-tat, always defect or win-stay lose-shift, and such labels now feed models of hybrid populations. Such a label claims something about the process behind a trajectory, yet the procedures that assign one answer for every input, whatever it contains. We put 4,800 agent-games from four models in the iterated prisoner’s dilemma through a read-out that asks deductively whether one canonical rule reproduces the whole trajectory, reports the set when several do, falls back on a classifier when none does, and abstains when that classifier is unsure. On synthetic play the deductive stage is correct 99.8% of the time and the abstention removes 59% of the residual error. On the corpus a rule is proved in 25.5% of agent-games, 92% of them AllC or AllD, whereas a *reactive* rule is proved in 2.0% and assigned in 30.3%, a gap robust to horizon controls. Nor is the mix a property of the agent: rescaling every payoff, which cannot change the game, moves it by up to 0.34 in total variation. Mining the unnamed quarter recovers two-phase play opening with a probe, and behind it drift. We propose a reporting standard for strategy labels.

Keywords: machine behaviour; large language models; iterated prisoner’s dilemma; strategy identification; selective classification; evolutionary game theory; multi-agent systems

1 Introduction

Large language models (LLMs) are increasingly deployed as *agents* that negotiate, coordinate and act on behalf of users [1, 2]. Their behaviour there emerges from strategic interaction rather than from isolated outputs, and is shaped by training, prompting, role assignment and linguistic framing, with direct implications for safety and governance. The proposal that machines be studied as behaving entities, with the observational tools of ethology and the experimental discipline of the behavioural sciences, has become considerably more urgent as the population of interacting artificial agents has grown [3].

The iterated prisoner’s dilemma is the natural testbed for that programme, and its appeal is specific: its strategy space is mapped. A two-by-two matrix with $T > R > P > S$ and $2R > T + S$ generates a landscape in which reciprocity, forgiveness, provocability and exploitation are defined precisely and their evolutionary fates computed [4–9], and it underpins work on the statistical physics of cooperation [10–12] and on mechanisms such as intention recognition and commitment

that displace the cooperative equilibrium [13, 14]. Because the map exists, importing it is irresistible, and a rapidly expanding literature now evaluates LLMs through game-theoretic lenses, reporting departures from Nash equilibria, cooperative biases, model-specific signatures and archetypes recovered from long-horizon play [15–19], alongside probes with the instruments of cognitive psychology [20] and models used as simulated economic subjects [21–23].

The output of that literature is, very often, a name. Model X plays tit-for-tat; model Y is a defector; model Z resembles win-stay lose-shift. A name of this kind is not a summary statistic. It is a claim about the process that produced a trajectory, and it licenses inferences that no rate licenses: that the agent will retaliate, that it will forgive, that it can be placed in a replicator equation and its fate computed. The claim is therefore worth exactly as much as the procedure that established it, and the procedures in use share a property that is rarely stated. A classifier returns its argmax for every input; a nearest-archetype distance returns a nearest archetype for every input. Neither can answer “none of these”, and neither distinguishes a trajectory a rule reproduces exactly from one that merely lies closer to that rule than to the others.

This paper takes strategy attribution itself as the object of study. We build a read-out with four possible verdicts rather than one, apply it to a fully crossed corpus of 2,400 dyads, and ask three questions.

1. How much of LLM play in the iterated prisoner’s dilemma can be shown, rather than guessed, to follow a canonical strategy, and which strategies survive that test?
2. Is the resulting distribution over strategies a property of the model, or does it move with features of the prompt that leave the game unchanged?
3. What is the play that the read-out declines to name, and is there a strategy in it that the canonical vocabulary is missing?

What makes the questions answerable is that the first stage of the read-out is deductive. All four canonical rules are memory-one, so asking whether a trajectory is exactly what AllC, AllD, tit-for-tat (TFT) or win-stay lose-shift (WSLS) would have played is a lookup, and only a unique match supports the statement that the agent played that rule. Several matching is an answer rather than a failure: an agent that cooperates for ten rounds against a cooperator *is* AllC and TFT and WSLS at once. Only where that stage returns nothing does a long short-term memory (LSTM) classifier [24] supply a nearest neighbour, and only when its posterior clears a floor. The four verdicts are reported separately throughout, never pooled.

Our contributions are: **(i)** a strategy read-out that separates deduction from inference and can abstain, validated on synthetic play where the generating rule is known and on a held-out rule that shows what an unseen strategy does to a classifier that cannot abstain (§3.1); **(ii)** a census showing that deduction reaches a quarter of the corpus, that 92% of that is constant play, and that reactive rules are labelled fifteen times more often than they are demonstrated (§3.2–§3.3); **(iii)** the finding that the strategy mix moves beyond a permutation null under every manipulation we apply, including a positive affine rescaling of the payoffs whose irrelevance is a theorem (§3.4); and **(iv)** a mining of the abstention set that recovers a two-phase, switch-once structure absent from the canonical vocabulary, together with the evidence that the rest of it is drift (§3.5–§3.7).

2 Methods

2.1 The game, the corpus and its manipulations

FAIRGAME (Framework for AI Agents Bias Recognition using Game Theory) [25] provides the computational infrastructure: conditions are defined in JSON, and at runtime FAIRGAME combines them with language-specific prompt templates, simulates repeated normal-form games and logs round-by-round trajectories. We adopt it and add payoff scaling. Taking its multilingual templates as given, rather than authoring our own, keeps the wording of the manipulations out of our hands.

Agents played a symmetric two-player prisoner’s dilemma (figure 1a). The templates state the cell values as *penalties*, years of imprisonment in the prisoner’s-dilemma narrative, and instruct the agent to *minimise* them, so a smaller number is a better outcome. Under that framing the base cells are $T = 0$, $R = 2$, $P = 6$, $S = 10$, satisfying $T < R < P < S$, which is the penalty-space image of the usual $T > R > P > S$. Option A is therefore the dominant, defecting action and Option B is cooperation. On the utility scale $u = -\text{penalty}$ the two normalised descriptors of the dilemma are greed $(T - R)/(T - S) = 0.20$ and fear $(P - S)/(T - S) = 0.40$.

We flag the polarity explicitly because inverting it would exchange the AllC and AllD columns of every table below. It is not a matter of interpretation: FAIRGAME logs the realised payoff alongside every action, and the assignment above reproduces the logged payoff on all 24,000 rounds of the corpus while the inverted assignment reproduces none of them.

Three factors were crossed on top of that fixed game (figure 1b). Every payoff cell was multiplied by a scale $\lambda \in \{0.1, 1, 10\}$; because $\lambda > 0$ this is a positive affine rescaling of a von Neumann–Morgenstern utility [26, 27], so the outcome ordering, the best-reply correspondence, the equilibrium set and the replicator dynamics are unchanged and the three games are one game presented with different numbers. The same game was instantiated in English, French, Vietnamese, Chinese and Arabic; a game is not changed by the language its rules are stated in. Each agent received a system-prompt persona, either *cooperative* or *selfish*, and the four ordered pairings were crossed with everything else; a persona is not irrelevant by theorem, but it carries no information about the game, since the four pairings present identical payoff matrices, and an agent was never told its opponent’s persona. The realised scale was recovered per game from the values logged rather than assumed from the directory.

The design crossed four models, Claude 3.5 Haiku, Gemini 3.5 Flash-Lite, GPT-4o and Mistral Large, with three payoff scales, five prompt languages, four persona pairings and ten replicates: $4 \times 3 \times 5 \times 4 \times 10 = 2,400$ dyads, 600 per model. Each dyad ran for ten rounds and both agents are recorded, giving 4,800 agent-games and 48,000 binary decisions, with exactly 40 dyads in every model $\times$ scale $\times$ language cell. Every decision resolves to one of the two option tokens and every dyad ran its full ten rounds, so no cell is thinned by a failure to parse. One GPT-4o dyad is logged with a mixed persona pairing where the design specifies a matched one, so that model’s persona margins are 599 and 601; no conclusion below turns on it.

On each round an agent received the full payoff table, its persona, the round index and the complete history of both players’ choices, and returned a single token naming one of two options. Communication was disabled, so the only channel between agents is the action history. Actions were labelled `OptionA` and `OptionB` in all languages, avoiding the connotations that “cooperate” and “defect” would carry; which label denotes cooperation was never stated and has to be read

off the payoff table.

Decoding used a fixed low output-token budget with reasoning traces disabled. This is not cosmetic: a model that emits hidden reasoning can exhaust a small budget inside the trace and return a truncated string, and a harness that falls back to a default action then produces an apparent cooperation rate of 1.0 that is an artefact of the parser. Parse-failure and fallback rates were logged for every cell and required to be below 0.02. Sampling temperature follows each provider’s recommended default, as in the FAIRGAME protocol: 1.0 for Claude, Gemini and GPT-4o, 0.3 for Mistral Large.

Two features of the design are confounded with model identity, and we declare them here rather than leave them to be discovered. The Gemini arm was run with the round count disclosed and the other three with the horizon hidden, so the Gemini column is a “model $\times$ horizon” column and is marked as such in figure 1b. Sampling temperature, fixed by the provider default above, likewise cannot be separated from model identity. Because both are constant within a model, neither can generate any within-model effect reported below; between-model comparisons carry them. The consequence we invoke wherever it bites is that no read-out statistic is comparable *across* models when decoding stochasticity would itself move the comparison, which includes the coverage figures of §3.2 and §3.5. The claims we draw there are about the panel, not about its ordering.

2.2 A read-out with four verdicts

A player’s trajectory is encoded as a sequence of two-letter tokens, $\langle \text{previous-round outcome} \rangle \langle \text{action this round} \rangle$, with outcome letters E (no history), R (both cooperated), S (the player cooperated and was exploited), T (the player exploited) and P (mutual defection). This is exactly the encoding of the synthetic corpus, which is what makes a classifier trained on generated play applicable to a transcript.

Stage one: deduction. Each canonical rule prescribes an action for every outcome letter: AllC cooperates everywhere, AllD defects everywhere, TFT plays C after E, R, T and D after S, P , and WSLS plays C after E, R, P and D after S, T . Asking whether a trajectory is exactly what rule k would have played is therefore a per-token lookup, checked without any learning. The matcher returns a *set*: exactly one member supports the statement that the agent played that rule, and several members is the correct answer when the realised history cannot separate them.

Stage two: inference, with the option of silence. Where no rule fits, an LSTM classifier trained on synthetic play from the four known generating rules returns a posterior over them, and we take its argmax as the *nearest* rule, never as an identification. The network is a single-layer LSTM (24-dimensional embedding, 96 hidden units) trained on the union of the noise-free and 5%-execution-noise splits of the ten-round synthetic corpus; nothing is retrained on LLM play. Its label is accepted only when the top posterior reaches a floor of 0.90, and otherwise the read-out abstains.

The floor is a convention here rather than a discovery. On thirty-round play the posterior-confidence distribution is strongly bimodal and 0.90 falls in the gap; at the ten-round horizon of this corpus it does not, so the value cannot be defended as a valley. We defend it instead by the

risk-coverage curve of §3.1, which we report in full, and give the census at floors of 0.80, 0.90 and 0.95 so that no conclusion rests on the cut.

Four verdicts therefore leave the read-out for every agent-game: *deduced* (exactly one rule reproduces the trajectory), *rule set* (several do), *nearest rule* (none does and the classifier is confident) and *abstained* (none does and the classifier is not). They are never pooled (figure 1c). We also record the minimum number of rounds on which play departs from its own nearest rule, which measures how much a nearest-neighbour label is being asked to absorb.

Grim trigger is a fifth memory-one rule and is treated asymmetrically, which we flag because the asymmetry runs against us: it enters every distance but not the exact-match vocabulary, so a trajectory only grim trigger would reproduce counts as unmatched. Admitting it raises the share of games matched by at least one rule by 0.0, 0.2, 3.3 and 0.6 percentage points for Claude, Gemini, GPT-4o and Mistral, and lowers Gemini’s uniquely matched share by 0.5 points. It is memory-one, (1, 0, 0, 0) opening with *C*, so it is inside the 32-rule class of §3.5 throughout.

2.3 Validating the read-out where the truth is known

The synthetic corpus is generated by replaying each of the four rules against sampled opponents, with execution noise at 0%, 5%, 10% and 20%; the network sees only the first two. Because the generating rule is recorded, three quantities that cannot be measured on LLM play are available there: whether the deductive stage recovers the truth when it fires, how accuracy grows with rounds observed, and what the abstention buys. We report the risk-coverage curve, the accuracy of the deductive stage conditional on firing, and the identifiability curve against its Bayes ceiling, the accuracy of an oracle that knows the generating distribution and is limited only by trajectories several rules produce alike.

A fourth quantity requires a rule the network has never seen. Generous tit-for-tat, which forgives a defection with fixed probability, was held out of training entirely and serves as a positive control for novelty: whatever the classifier does with it is what it will do with any strategy outside its vocabulary.

2.4 Widening the vocabulary, and the nulls

A rule may fail to fit because the vocabulary is too narrow rather than because the play has no rule in it, so the abstention set is mined with three progressively wider hypothesis classes, each paired with a null. The first is all 32 deterministic memory-one rules, an opening move crossed with an action for each of the four conditioning states. The second admits trajectories built from two such rules with a single switch. The third is a library of fifteen named non-canonical strategies from the tournament literature, including suspicious TFT, grim trigger, tit-for-two-tats, gradual, prober, contrite TFT, soft and hard majority and the alternator, together with the two-phase clock rules spliced from an ordered pair of opponent-only strategies. Only AllC, AllD and TFT can be spliced this way, since WSLs reads the player’s own last action, which gives six ordered pairs at nine switch points and 69 candidate rules in all. That count is why the null matters: with 69 chances to fit a ten-round binary sequence, a coverage figure means nothing until the same search has been run on shuffled play.

Every coverage figure is paired with a permutation null in which the focal player’s own action sequence is shuffled in place, holding its cooperation rate and the opponent’s realised sequence fixed. The null is not a formality. A two-segment fit over ten rounds has enough freedom to

reproduce a large share of shuffled play by chance, and constant play is invariant under the shuffle by construction, so raw coverage on its own is not evidence of structure.

Two further probes ask what the play is without assuming a vocabulary at all. The first counts every four-round window of the focal player’s own actions and compares its frequency with the same within-game shuffle, so a lift above one is temporal structure and nothing else; alternation is the pattern a memory-one rule cannot express, so the CDCD and DCDC windows are the ones to read first. The second scores nested predictors of the next action, memoryless, memory-one, memory-two and a memory-one rule allowed to change with the phase of the game, by the Bayesian information criterion [28] on identical rows, so the comparison is about history rather than sample size.

Finally, we ask whether the unnameable play is a strategy at all. A trajectory generated by a fixed rule is coherent across a game by construction, so we split each game at round five and correlate the cooperation rate of the two halves within each verdict class, and we cluster the abstention set in memory-one profile space, sweeping k from 2 to 8 and reporting silhouette alongside a bootstrap adjusted Rand index.

2.5 Statistical analysis

All intervals are 95% bootstrap confidence intervals over 2,000 resamples that resample whole dyads [29]. This is consequential: rounds within a game are autocorrelated and the two agent-rows of a dyad share a history, so a row-level resample would understate the interval by roughly $\sqrt{2}$.

Whether a manipulation moves the strategy mix is answered with a distributional test rather than a contrast on a single share. For each model and factor we take the largest total-variation distance between the five-category mixes of any two levels, and compare it with a permutation null that shuffles the factor label across *dyads*, so a dyad keeps both of its agent-rows together; 2,000 permutations are drawn. We report the observed distance, the permutation P value and the null’s 95th percentile, which doubles as the smallest shift the test could have detected.

We do not correct for multiple comparisons and say so rather than leave it to be inferred. Each model is a separate system and is treated as its own family, so the tests within a manipulation are four families of one rather than one family of four; the claims we draw are in any case about whether manipulations move the mix at all, not about which model moves most.

3 Results

3.1 What the read-out is worth

The deductive stage is exact by construction, but exactness is not usefulness: it could fire so rarely as to be decorative, or so often as to be uninformative. On the synthetic test set it fires on 80.3% of trajectories with a unique answer and recovers the generating rule in 99.8% of those; a further 0.7% receive a set, which contains the truth 99.4% of the time; the remaining 19.0% match nothing, which is what 5% execution noise does to an exact matcher over ten rounds. Deduction is therefore both frequent and near-certain when the play really is rule-generated. This is the benchmark against which the LLM figures of §3.2 should be read.

Crucially, the deductive stage is not biased towards the constant rules at this horizon. Resolved by generating rule, it returns a unique answer on 79.7% of AllC-generated trajectories,

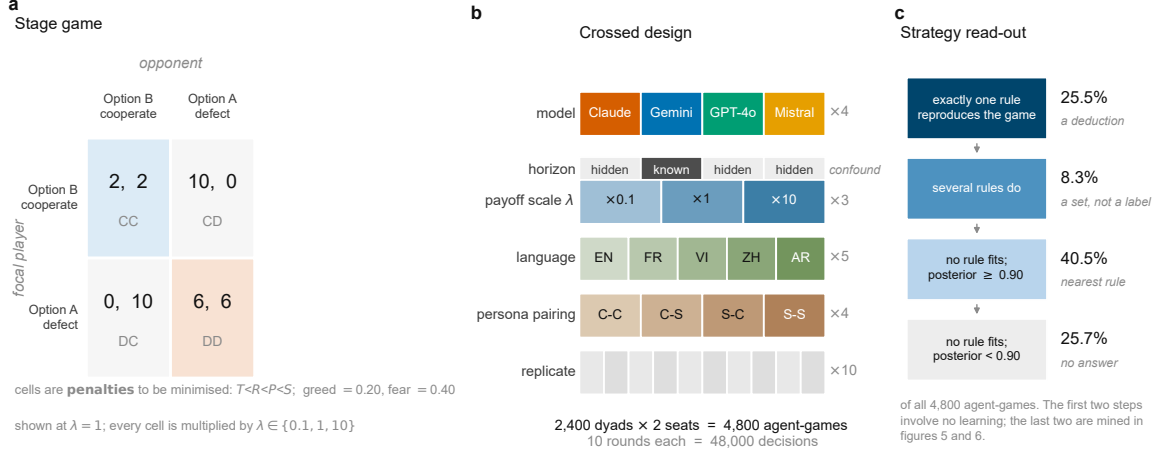


Figure 1: The game, the design and the read-out. (a) The stage game at $\lambda = 1$. Cells are penalties the agent is told to minimise, so $T < R < P < S$ and Option A is the dominant, defecting action. Greed and fear are computed on the utility scale $u = -\text{penalty}$. Every cell is multiplied by $\lambda \in \{0.1, 1, 10\}$, a positive affine rescaling that leaves the game strategically identical. (b) The fully crossed design: 2,400 dyads, 4,800 agent-games, 48,000 binary decisions, with 40 dyads in every model $\times$ scale $\times$ language cell. The horizon row is shaded as a confound rather than a factor, because only the Gemini arm was told the round count. (c) The read-out and what it returns on this corpus. The first two steps are deductive lookups over the four canonical memory-one rules and involve no learning; the third is an LSTM trained on synthetic play, admitted only above a posterior floor; the fourth is an abstention. Shares are of all 4,800 agent-games.

80.9% of AllD, 80.8% of TFT and 79.6% of WSLS, and where it fires it is right 99.8%, 100.0%, 99.6% and 99.7% of the time. Ten rounds are enough to prove a reactive rule as often as a constant one, which forecloses the reading that §3.3 is a statement about the horizon rather than about the play.

The classifier that covers the remainder needs ten rounds and no more (figure 2a). Accuracy rises from 0.487 after one round, where the opening move alone separates AllD from the three rules that open cooperatively, to 0.984 after ten, against a Bayes ceiling of 0.993. The gap to the ceiling is 0.009. Whatever the read-out fails to do on LLM play is therefore not a failure of horizon or of capacity on this vocabulary. Under execution noise the classifier degrades gracefully within its training range, 0.995 at 0% and 0.973 at 5%, and less gracefully outside it, 0.937 at 10% and 0.810 at 20%.

That degradation is what the abstention is for (figure 2b). At the 0.90 floor, the read-out declines to label 2.7% of the low-noise test set and error among the games it still labels falls from 1.65% to 0.43%. On the unseen 10%-noise split it declines 9.1% of games and error falls from 6.34% to 2.61%, a 59% reduction. The floor is a chosen operating point on a curve rather than a threshold with independent standing, which is why we report the curve and repeat the census below at 0.80 and 0.95.

The held-out rule makes the sharpest case for having an abstention at all (figure 2c). Generous tit-for-tat was never shown to the network. It is absorbed: 90.1% of its games are labelled TFT, 5.1% WSLS, 4.5% AllC and 0.3% AllD. Nothing in the output says “this is not one of my four”. A read-out that must return a label will therefore report a vocabulary it was given rather than a strategy the agent played, and it will do so with no external sign that anything is wrong. This is not a defect of the particular network; it is what forcing an argmax means.

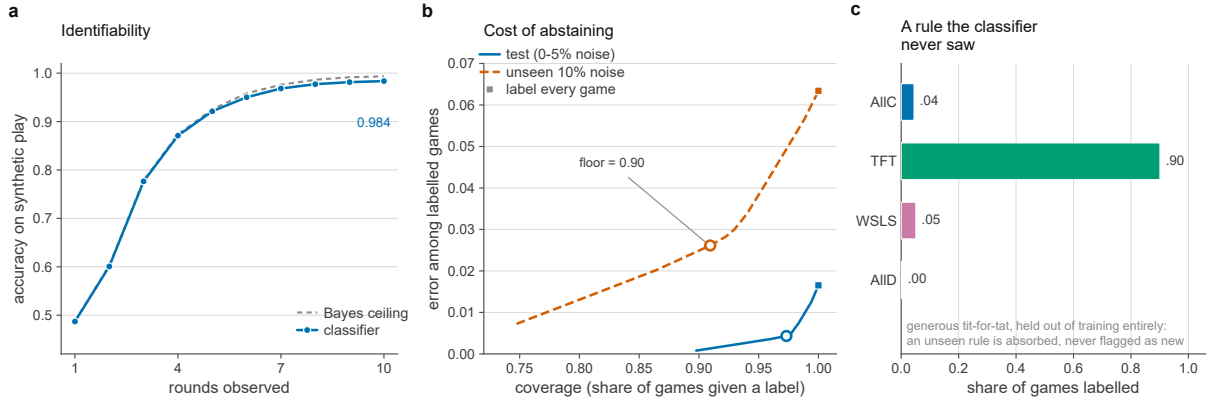


Figure 2: Validation on synthetic play, where the generating rule is known. (a) Accuracy as a function of the number of rounds observed, against the Bayes ceiling of an oracle that knows the generating distribution and is limited only by trajectories several rules produce alike. Ten rounds are enough for the classifier to reach 0.984 against a ceiling of 0.993, so what fails on LLM play in figures 3–6 is not the horizon. (b) The risk-coverage curve. Sweeping the posterior floor trades coverage for error among the games still labelled; squares mark labelling every game and rings mark the 0.90 floor used throughout. On play 10% of whose actions are noise, a rate the network never saw in training, abstaining on 9% of games removes 59% of the error. (c) What a classifier does with a rule it has never seen. Generous tit-for-tat was held out of training entirely; 90% of its games are labelled TFT and none is flagged as new. Any read-out without an abstention will absorb an unknown strategy into its vocabulary rather than report one.

3.2 A census of named play

Applied to the corpus, the read-out deduces a unique rule in 25.5% of agent-games, returns a set in 8.3%, supplies a nearest rule in 40.5% and abstains in 25.7% (figure 3a). The spread across models is large: Mistral Large is deduced in 48.7% of its games and Claude 3.5 Haiku in 9.7%, with Gemini 3.5 Flash-Lite and GPT-4o between at 21.9% and 21.8%. Because sampling temperature is confounded with model identity (§2.1) we draw no conclusion from that ordering. The panel-level statement is what matters and it does not depend on the ordering: in every model at least half of the play is reproduced by no canonical rule at all, from 50% for Mistral Large to 86% for Claude 3.5 Haiku.

The census is stable against the abstention floor in the direction that matters. Moving the floor from 0.80 to 0.90 to 0.95 moves the abstained share from 20.0% to 25.7% to 32.4% and moves nothing else, because the deductive stage runs first and is unaffected: the deduced and rule-set shares are 25.5% and 8.3% at every floor. No claim below depends on where inside that band the cut is placed.

The composition of the deduced quarter is the first substantive result. Of the 1,224 agent-games in which exactly one canonical rule reproduces the whole trajectory, 1,129, or 92.2%, are AllC or AllD. The share of the corpus in which a *reactive* rule, TFT or WSLs, is deductively established is 95 agent-games, 2.0%. Per model the constant share of the deduced set is 68.1% for Claude, 90.1% for Gemini, 90.4% for GPT-4o and 98.8% for Mistral. What the deductive stage mostly proves, in other words, is that an agent did the same thing ten times. The rules that carry the theoretical content of the iterated prisoner’s dilemma, the ones that condition on what the opponent did, are almost never proved.

One reading of that asymmetry has to be excluded before it can be believed. A reactive rule

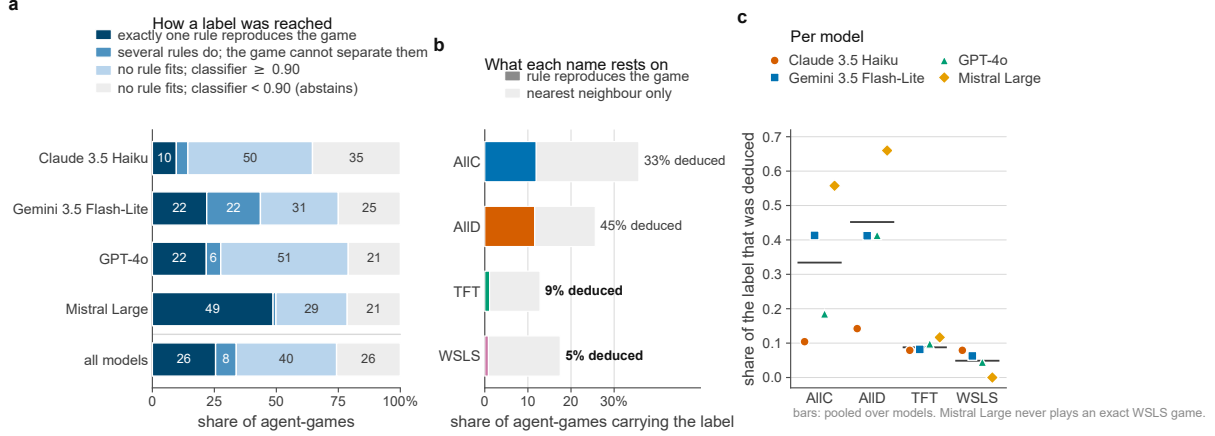


Figure 3: What a strategy label is made of. (a) The four verdicts per model and pooled. Dark: exactly one canonical rule reproduces the whole trajectory. Mid: several do, and the ten rounds do not separate them. Pale: none does, and the classifier supplies a nearest rule. Lightest: none does and the classifier abstains. (b) The same corpus resolved by label rather than by game. Each bar is the share of agent-games carrying that name, split into the part a rule reproduces exactly and the part that is a nearest-neighbour statement. AllC and AllD are largely deduced; TFT and WSLS are almost entirely inferred. (c) The same quantity per model. Bars mark the pooled value. Mistral Large is assigned WSLS in 14.3% of its agent-games and plays an exact WSLS game in none of them.

is only distinguishable from a constant one when the opponent gives it something to react to: against an opponent that never defects, TFT and AllC prescribe the same ten actions and no method can separate them. We therefore restrict to the agent-games whose opponent history affords the distinction, playing at least one *C* and at least one *D* in rounds one to nine. That is 66.1% of the corpus, ranging from 50.1% for Mistral Large to 88.4% for Claude. Inside it the deduced-reactive share rises only from 2.0% to 2.5%, while the deduced-constant share is 18.0%. The scarcity of proved reactive play is therefore not an artefact of opponents who never gave these agents an occasion to reciprocate: it survives conditioning on exactly those occasions, at a horizon where the same matcher proves TFT on four fifths of TFT-generated play (§3.1).

That composition also explains a result that would otherwise look damning for the deductive stage. Against the shuffled null of §2.4, canonical coverage is barely enriched: the excess is 1.8 percentage points for Claude, 1.5 for Gemini, 2.0 for GPT-4o and 0.4 for Mistral. It could not be otherwise, because a constant trajectory is invariant under a permutation of its own actions, so every AllC and AllD match survives the shuffle by construction and those are 92% of the matches. The enrichment test is therefore uninformative about constant play and informative about reactive play, where it says reactive matches are rare in the data and in the null alike.

3.3 The labels that do not survive the audit

Resolving the same corpus by label rather than by game is where the two stages come apart (figure 3b). AllC is assigned to 35.7% of agent-games and 33.4% of those assignments are deductions; AllD is assigned to 25.6% with 45.2% deduced. TFT is assigned to 12.8% of agent-games with 8.8% deduced, and WSLS to 17.5% with 4.9% deduced. Put the other way, 91.2% of TFT labels and 95.1% of WSLS labels in this corpus are nearest-neighbour statements about play that no rule in the vocabulary reproduces.

The gap is not a rounding artefact of a small denominator. Reactive labels are handed out to

Table 1: The read-out per model. Every model contributes 600 dyads, 1,200 agent-games and 12,000 decisions. “Deduced” is the share of agent-games in which exactly one of the four canonical rules reproduces the whole trajectory; “constant” is the share of that deduced set which is AllC or AllD. “Reactive” compares the share of agent-games in which TFT or WSLS is deductively established with the share carrying one of those two labels by any route. “ λ shift” is the largest total-variation distance between the strategy mixes at two payoff scales, whose null is exactly zero because the rescaling is a positive affine transformation; the null’s 95th percentile is given for scale. The last column mines the abstained games with the extended vocabulary of §2.4 against its shuffled null. Coverage statistics are not comparable across models, because sampling temperature is confounded with model identity (§2.1).

Model	Coop.		constant	Reactive rule		Abstained	Abstained set named	
	rate	Deduced		proved	labelled		obs.	null
Claude 3.5 Haiku	0.63	9.7%	68.1%	3.1%	39.0%	35.4%	0.27	0.10
Gemini 3.5 Flash-Lite	0.74	21.9%	90.1%	2.2%	30.5%	25.0%	0.56	0.14
GPT-4o	0.45	21.8%	90.4%	2.1%	32.4%	21.0%	0.39	0.07
Mistral Large	0.52	48.7%	98.8%	0.6%	19.3%	21.4%	0.59	0.10
All models	0.58	25.5%	92.2%	2.0%	30.3%	25.7%	0.43	0.10

Every λ shift exceeds its permutation null: 0.263, 0.338, 0.325 and 0.125 against null 95th percentiles of 0.113, 0.123, 0.105 and 0.098 ($P < 0.001$ for the first three, $P = 0.006$ for Mistral Large).

30.3% of agent-games and reactive rules are demonstrated in 2.0%, a factor of 15.3. Per model the ratio is 12.6 for Claude, 14.1 for Gemini, 15.6 for GPT-4o and 33.1 for Mistral Large, whose WSLS label appears in 14.3% of its agent-games and is deduced in none of its 1,200 (figure 3c). A statement that Mistral Large plays win-stay lose-shift is, on this corpus, a statement that its play was closer to WSLS than to the other three, in games none of which WSLS would have produced.

What that proximity is worth can be quantified. Where a label is assigned by nearest neighbour, play departs from the rule it names on 1.97 rounds out of ten for AllC, 2.01 for AllD, 2.49 for TFT and 2.09 for WSLS. A rule that is wrong on a fifth to a quarter of the decisions it is supposed to generate is not a description of the process; it is the closest of four available descriptions, three of which are also wrong.

Table 1 collects the per-model read-out together with the two quantities the rest of the paper turns on, the shift in the mix under payoff rescaling and the coverage the wider vocabulary buys on the abstained games.

This is a claim about warrant, not about the network. The classifier is accurate where accuracy is definable (§3.1), and its nearest-neighbour statements are not errors, since there is no truth for it to be wrong about. The error is in reading one as an identification, which a single pooled label distribution invites and reporting the four verdicts separately prevents.

3.4 The strategy mix is a property of the prompt

A strategy label is meant to name something about the agent. If the distribution over labels is instead a function of how the game was written down, a census measures the prompt.

It is (figure 4). Every one of the sixteen model $\times$ factor comparisons moves the strategy mix beyond its permutation null. The sharpest case is the payoff scale, because there the null is not a convenient default but a theorem: multiplying every cell by $\lambda > 0$ leaves best replies, equilibria and replicator dynamics unchanged, so a strategic agent’s mix must not move at all. It moves

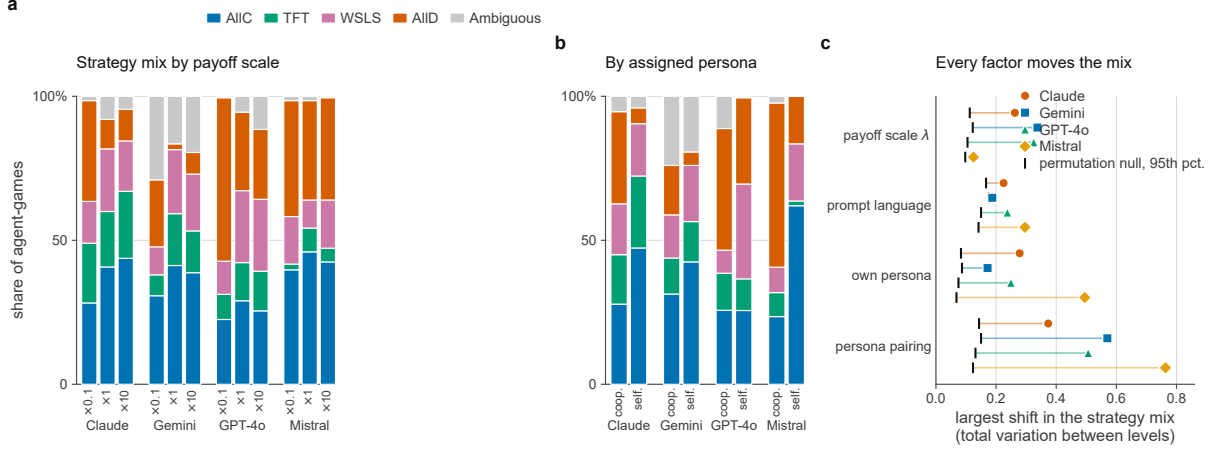


Figure 4: The distribution over strategies moves with manipulations that leave the game unchanged. (a) The strategy mix at each payoff scale, per model. Multiplying every cell by a positive constant is a positive affine rescaling of a von Neumann–Morgenstern utility, so the three games are one game and the predicted shift is exactly zero. GPT-4o’s AllD share falls from 0.57 to 0.24 across it. (b) The same by assigned persona. Mistral Large told it is cooperative plays AllD in 57% of its games and AllC in 24%; told it is selfish, 17% and 62%. The instruction moves the mix opposite to its own semantic content. (c) Whether the mix moves at all, for every model and factor. Markers give the largest total-variation distance between the mixes at two levels; ticks give the 95th percentile of a permutation null that shuffles the factor label across whole dyads. All sixteen comparisons exceed their null.

by 0.263 for Claude, 0.338 for Gemini, 0.325 for GPT-4o and 0.125 for Mistral in total variation, against null 95th percentiles of 0.113, 0.123, 0.105 and 0.098 respectively, with $P < 0.001$ for the first three and $P = 0.006$ for the fourth. Resolved into shares, GPT-4o is labelled AllD in 56.8% of its games at $\lambda = 0.1$ and in 27.2% at $\lambda = 1$ and 24.2% at $\lambda = 10$; Claude’s AllD share falls from 35.0% to 10.2% and Gemini’s from 23.2% to 2.0% across the same rescaling. Writing the same payoffs as 0.2 rather than 2 roughly halves the share of play that is recorded as unconditional defection.

Because “Ambiguous” is a statement about identifiability rather than about behaviour, a shift in the five-category mix could in principle be games moving into and out of that bin rather than a change in what the agents do. It is not. Repeating every test over the four named strategies alone leaves all sixteen comparisons above their nulls, with the payoff-scale distances at 0.245, 0.304, 0.296 and 0.127 and no P value above 0.0065.

Prompt language moves the mix in all four models as well, by 0.188 to 0.296 against nulls of 0.142 to 0.175. Here the null is a weaker claim than for the payoff scale, and one caveat is a finding rather than a nuisance: the FAIRGAME templates do not state the objective identically in every language, since English, French and Vietnamese instruct the agent to minimise a penalty while Chinese and Arabic instruct it to maximise a reward, so a five-language comparison mixes a language manipulation with an objective-framing manipulation. We report the five-language test as the omnibus it is and do not decompose it, because the strategy mix is the dependent variable here and the framing subset leaves too few dyads per cell to estimate a five-category distribution stably.

The weakest of the sixteen comparisons is Gemini’s language test, at 0.188 against a null 95th percentile of 0.175 ($P = 0.028$), and it is the one entry that a family-wise correction across the four models would absorb. It is also the one entry that the four-category re-test strengthens

rather than weakens, to 0.205 at $P = 0.003$.

The persona manipulation produces the largest shifts in the study, and it is the most interesting because it runs backwards. Mistral Large told it is cooperative is labelled AllD in 57.0% of its games and AllC in 23.5%; told it is selfish, the same model is labelled AllD in 16.5% and AllC in 62.0%. Claude moves from 32.0% AllD and 27.8% AllC to 5.5% and 47.3%. In every model the selfish persona produces less unconditional defection than the cooperative one, and in three of the four it produces more unconditional cooperation; GPT-4o is the exception only in that its AllC share is flat at 0.26 while its AllD share falls from 0.42 to 0.30. Resolving the manipulation into the four ordered pairings moves the mix by 0.373 to 0.763, against own-persona shifts of 0.172 to 0.495. An agent is never told its opponent’s persona, so the excess of the pairing over the own-persona effect is a response to the interaction rather than to the instruction.

Both directions are consistent with a single reading, which we offer as the most parsimonious account rather than as a demonstration: that the models treat the displayed numbers as rewards to be maximised rather than as penalties to be minimised, pattern-matching the matrix to the canonical reward-framed dilemma in which the largest entry is the prize. Under that reading Option A yields the symmetric pair (6, 6) and Option B can yield the single largest number, 10, so an agent reaching for a mutually good outcome takes A and one reaching for the largest attainable number takes B, which is what the persona columns show; and enlarging λ makes that maximum more conspicuous, which is the direction the scale panel shows. Separating it from an option-ordering account requires restating the identical matrix in explicit reward form, an experiment we did not run.

Whichever account is right, the consequence for attribution is the same and does not depend on it. A strategy census is not a measurement of the agent unless it is invariant under transformations that leave the game alone, and on this corpus it is not.

3.5 The anatomy of what the read-out will not name

A quarter of the corpus leaves the read-out with no label. The easy dismissal is that this residual is noise; answering it decides whether the vocabulary is too small or the wrong shape.

It is not a shrug (figure 5a). The posterior in the abstained games is not spread over four labels but split between two: the top pair is AllC versus TFT in 29.9% of cases, TFT versus WSLS in 19.0%, WSLS versus AllD in 18.9%, AllC versus WSLS in 17.2% and TFT versus AllD in 14.3%, with AllC versus AllD, the only pair with no rule between them, at 0.6%. TFT appears in 63.2% of the ties. The network is refusing to choose between adjacent rules, not failing to see anything.

Nor is the play near the vocabulary (figure 5b). In the reactive square, where AllC, TFT and AllD are corners, mean distance to the nearest corner is 0.024 for deduced games, 0.333 for those given a nearest rule and 0.417 for those abstained, with medians of 0.000, 0.320 and 0.471 on a square whose corners are at most 1.41 apart. The ordering is the point: a nearest-neighbour label is assigned to play a third of the way to the nearest canonical point, and the games the read-out abstains on are further still.

The motif analysis says what kind of structure it is (figure 5c), and it contradicts the most natural guess. A plausible hypothesis for unnameable play in a two-action game is alternation, since a strictly alternating sequence is invisible to a memory-one rule. It is not what is there. Relative to a within-game shuffle that preserves each player’s own cooperation rate, the four-

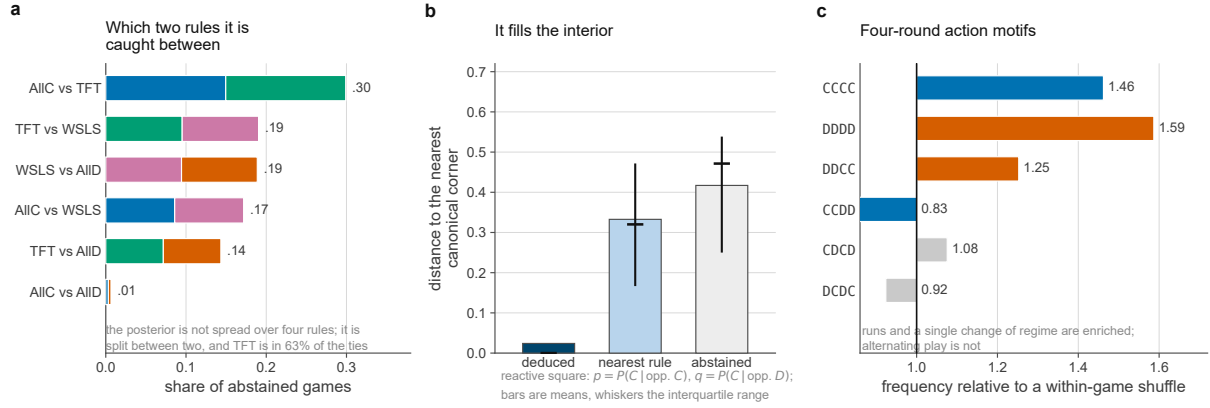


Figure 5: The abstention set is structured, and the structure is outside the vocabulary. (a) Which two rules the posterior is split between. Each bar is halved between the pair. The network is not spreading mass over four labels; it is caught between two, and TFT is in 63% of the ties. (b) Distance to the nearest canonical corner of the reactive square $p = P(C | \text{opponent } C)$, $q = P(C | \text{opponent } D)$, in which AllC, TFT and AllD are corners. Bars are means and whiskers the interquartile range. Deduced play sits on a corner by construction; abstained play sits in the interior. (c) Frequency of four-round windows of the focal player’s own actions, relative to a shuffle of that player’s own actions within its own game. Runs and a single change of regime are enriched; alternating play, the pattern a memory-one rule cannot express, is not.

round windows enriched in the abstention set are the runs and the single switch: DDDD at a lift of 1.59, CCCC at 1.46 and DDCC at 1.25, while CDCC sits at 1.08 and DCDC at 0.92, which is chance. Only 9.4% of abstained games alternate more than their own base rate implies by a margin of 0.15 or more, and pooled over the set the alternation excess is negative, -0.036 ($[-0.046, -0.027]$): these agents persevere. Whatever the abstention set is made of, it is made of long runs with a change of regime in them.

3.6 Two-phase play, opening with a probe

Widening the vocabulary answers the question the canonical four cannot (figure 6a). The 69 candidates of §2.4 reproduce 43.3% of the abstained games exactly, against a shuffled null of 10.1%, an excess of 33.2 percentage points. On the games the classifier was confident about, the same vocabulary reproduces 34.4% against a null of 13.3%. Per model the abstained coverage is 27.3% for Claude, 55.7% for Gemini, 39.3% for GPT-4o and 59.1% for Mistral, against nulls of 9.6%, 14.0%, 6.7% and 9.7% in the same order. For 90.2% of the abstained games the nearest rule in the wide library is not one of the canonical four.

What lands is specific (figure 6b). Of the 534 abstained games the extended vocabulary reproduces exactly, 66.9% are two-phase rules with a single switch and the median switch falls at round two. The single most frequent fit is AllD followed by AllC, at 39.1% of the exact fits; then suspicious TFT, which opens with defection and reciprocates thereafter, at 18.7%; then TFT followed by AllD at 18.7%, TFT followed by AllC at 4.7%, grim trigger at 3.7%, the alternator at 2.6%, AllC followed by AllD at 2.4% and prober at 1.9%.

Two features of that list are worth separating from the arithmetic. The first is its direction: 65.5% of the fitted switches end in a more cooperative regime than they began, against 31.7% that end in a less cooperative one and 2.8% that end in TFT. Backward induction on a finite horizon predicts the opposite, and the switches do not fall where an endgame effect would put

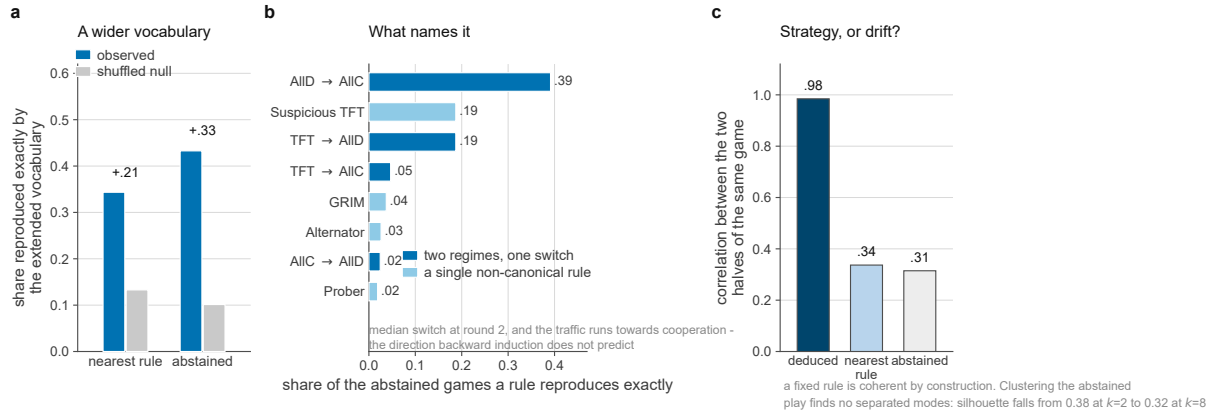


Figure 6: What the abstained play is, and whether it is a strategy. (a) Share of games reproduced exactly by the extended vocabulary of fifteen named non-canonical rules plus 54 two-phase clock rules, against the permutation null that shuffles the focal player’s own actions. The wider vocabulary earns 0.33 of coverage above chance on the abstained games and 0.21 on those given a nearest rule. (b) The rules that reproduce an abstained game exactly. Two thirds of the fits are two-phase, the median switch falls at round two, and the traffic runs towards cooperation. (c) Correlation between the cooperation rate of the first and second halves of the same game. Play that a rule reproduces is coherent by construction; the abstained set is barely less coherent than play the classifier labelled confidently, and clustering it finds no separated modes.

them, but at a median of round two. Whatever produces them is an opening phenomenon.

The second is that three of the four most frequent fits share a shape: an opening defection that is not sustained. AIID-then-AIIC, suspicious TFT and prober all begin by defecting and then hand control to something else, and together they account for 59.7% of the exact fits. Read with the persona inversion of §3.4, the natural description is a probe: the opening move is used to sample the opponent rather than to commit to a line of play, and the strategy proper starts on round two or three. That shape is not in the canonical vocabulary at any width, because it is not memory-one; it is a rule about the round index. It is also not what any of the standard archetypes would predict, and it is the substantive answer to what LLM play in this corpus consists of when it is not constant.

The mechanism behind the failure of the memory-one class is visible directly. Scoring nested predictors of the next action on identical rows, all four models select memory-two by BIC: the improvement over a memoryless baseline goes from -1685 to -2399 for Claude, -4334 to -4792 for Gemini, -3684 to -4865 for GPT-4o and -6652 to -9158 for Mistral. Allowing the memory-one rule to change with the phase of the game instead, an equally sized extension in the direction of time rather than of history, is worse than memory-two for every model. The rows are pooled within a model, and a heterogeneous population of memory-one agents can look memory-two when pooled, so this is evidence that the corpus is not memory-one rather than that any single agent-game is.

One limit applies to the wider vocabulary itself and bounds how much of this can be read as identification. Over ten rounds a trajectory rarely visits all four conditioning states, so rules that differ only in an unvisited state cannot be told apart. Of the 4,800 agent-games, 2,608 match none of the 32 deterministic memory-one rules, and of those that do match, most are matched by four or eight at once; exactly 89, or 1.9% of the corpus, are pinned to a single one. A coverage figure for the wider class is a statement about a *set* of rules and not about a rule,

and the ten-round horizon is what makes it so.

3.7 Strategy, or drift

Whether the unnameable play is a strategy the vocabulary is missing or no strategy at all has an answer the data can bound, and it is both (figure 6c).

A trajectory generated by a fixed rule is coherent across the game by construction, and the deduced games confirm the instrument: the correlation between the cooperation rates of the two halves of a game is 0.985 there. It is 0.337 among games given a nearest rule and 0.315 among abstained games. The first half of an abstained game barely predicts its second, and, importantly, neither does the first half of a game the classifier was confident about. This is not a property of the abstention set alone; it is a property of all LLM play in this corpus that no rule reproduces.

Clustering finds no fifth archetype. In memory-one profile space the silhouette of a k -means partition of the abstention set falls from 0.378 at $k = 2$ to 0.319 at $k = 8$ and never rises above its value at $k = 2$, so there is no number of clusters the data prefer. The bootstrap adjusted Rand index is high at small k , 0.93 at $k = 2$ and 0.94 at $k = 3$, but that measures whether a partition is reproducible rather than whether it is separated, and a single elongated cloud cut in half reproduces perfectly. The three-cluster solution is interpretable, with one mode that persists and imitates (self-persistence 0.80, opponent-matching 0.82, 56% of the set), one near 0.5 on every coordinate (31%) and one that is anti-imitative (opponent-matching 0.12, 13%), but these are regions of a continuum rather than modes.

The verdict is therefore split, and we state it as such. About 43% of the abstention set is a rule outside the canonical four, most often a two-phase rule with an opening probe, and that share is more than four times its shuffled null. The rest of it, and the majority of the play the classifier labels confidently, is within-game drift that no fixed rule will name, because there is no fixed rule there to find.

4 Discussion

4.1 A label needs a provenance

The central result is a gap between two quantities that are usually reported as one. Reactive strategy labels are assigned to 30.3% of the agent-games in this corpus and reactive rules are deductively established in 2.0% of them. Both numbers are correct; they answer different questions, and only the second supports the inferences that a strategy name is normally used to license.

The distinction is not pedantic, because the two behave differently under everything else in the paper. The deduced share is unaffected by where the abstention floor is placed, while the inferred share is not. The deduced share is dominated by constant play, so it is nearly invariant under a permutation of a player’s own actions, which is why a null of that kind tests the reactive part of the vocabulary and nothing else. And the deduced share is the only one for which the validation of §3.1 transfers, because it is the only one for which “correct” is defined without reference to a training distribution.

The remedy is cheap: an exact matcher costs nothing next to a classifier, and reporting its output as a separate column costs nothing at all. We propose that any strategy attribution for

an artificial agent carry three shares, deduced, several-rules-fit and nearest-neighbour, together with the mean number of rounds on which play departs from the rule it was assigned. On this corpus those are 25.5%, 8.3%, 66.2% and 1.97 to 2.49 rounds out of ten.

4.2 The vocabulary is the wrong shape, not the wrong size

The natural repair for a vocabulary that does not fit is to enlarge it, and the way that repair fails here is informative. Going from four rules to all 32 deterministic memory-one rules and then to two of them in sequence leaves a substantial residual, and the wide library that does earn 33 points over its null earns them mostly with rules that are not memory-one at all but two-phase: a regime for the opening and a regime for the rest, with a median switch at round two. The BIC comparison points the same way, selecting memory-two over memory-one for every model, and the motif analysis rules out the obvious alternative, since alternation is at chance while runs and single switches are enriched.

Together these say the class is misspecified rather than under-parameterised. A memory-one description asks what an agent does after each joint outcome; these agents are doing something that depends on where in the game they are as well as on what just happened, concentrated in the first two or three rounds. A vocabulary for LLM agents should therefore be built from phases and probes rather than by adding rules to a memory-one enumeration.

The direction of the switches sharpens the point. Two thirds run towards cooperation and they cluster at the opening, not the endgame. The canonical prediction for a finite horizon is unravelling towards mutual defection; what this corpus contains is an opening defection that is abandoned. That is closer to exploration than to backward induction, and a population model assigning these agents a fixed strategy would get both the sign and the timing wrong.

4.3 Consequences for evolutionary models of hybrid populations

Efforts to model populations in which artificial and human agents interact [2, 30, 31] typically assign agents strategies from a known family and compute the resulting dynamics. Three of our results bear on that programme. First, the assignment is mostly not available: a replicator equation needs each agent placed at a point, and here that point is deductively determined for a quarter of agent-games and is a constant rule in 92% of those. Second, it is not stable under transformations the dynamics cannot see, since rescaling all payoffs leaves the replicator equation identical and moves the mix by up to 0.34, so two population models a game theorist would call the same model predict differently about the same agent. Third, the individual game is not coherent enough to carry a label at all, with a between-half correlation of 0.32 outside the deduced set. What these support is a distribution over strategies whose support changes with framing, not a strategy.

4.4 Relation to the strategy-attribution literature

The closest neighbours are studies that recover archetypes from LLM play: Fontana et al. [17] over long horizons, Akata et al. [15] as stable model-specific signatures, and Lorè and Heydari [19], who separate game structure from contextual framing and find framing carries comparable weight. Our results are not in tension with those measurements, and we do not claim their labels are wrong. What we add is the denominator: an archetype recovered from LLM play

is a nearest-neighbour statement unless a deductive stage says otherwise, and on our corpus it says otherwise for 2.0% of reactive labels. The framing sensitivity that Lorè and Heydari [19] document for outcomes we document one level up, for the strategy distribution itself, under a manipulation whose null is a theorem rather than a default. That also connects the machine-behaviour programme [3] to the framing literature in human decision making [32, 33], with an instructive disanalogy: human framing effects map to a compact set of psychological mechanisms, whereas what moves here is the identity of the strategy a standard read-out reports, and it moves under a change of units.

4.5 Limitations

Four limitations bound these conclusions, and the first two are the ones that would change the numbers rather than the interpretation.

The horizon is ten rounds, and this bounds §3.6 in particular. It under-identifies the memory-one vocabulary, so most games matched by the wider class are matched by a set of rules rather than by one and only 1.9% of the corpus is pinned to a single one of the 32; it gives a two-segment fit enough freedom that a shuffled null already reaches 0.10, which is why every coverage figure is reported against that null; and it compresses any endgame effect, so the absence of endgame switching is weaker evidence than the presence of opening switching. It does not, however, bound §3.3, and the two controls that establish this are the ones we would put to a sceptical reader first: at ten rounds the same matcher pins a unique answer on 80.8% of TFT-generated and 79.6% of WSLS-generated synthetic play, as often as on the constant rules, and the deduced-reactive share barely moves when the corpus is restricted to the two thirds of agent-games whose opponent history affords the distinction.

The model set is four, a small sample of a fast-moving population, and nothing here establishes that the pattern generalises to models with explicit reasoning procedures. Testing the read-out on reasoning models is the single most valuable extension, and we regard these results as establishing the instrument rather than settling the question.

Two confounds with model identity remain. The Gemini arm knew the round count and the others did not, so that column mixes model identity with horizon knowledge; sampling temperature is likewise fixed by provider default, so no between-model ordering of a coverage statistic is interpretable and none is claimed. Neither is constant across the levels of any manipulation, so neither can generate the within-model effects of §3.4.

Finally, the abstention floor is a convention at this horizon rather than a valley in the posterior, and the reward-reading account of the persona and scale effects (§3.4) is an inference from converging evidence rather than a controlled demonstration. We report the risk-coverage curve and the census at three floors for the first, and identify the relabelling experiment that would settle the second.

4.6 Conclusion

Four language models played 2,400 iterated prisoner’s dilemmas, and a read-out that separates deduction from inference and is allowed to abstain establishes a canonical strategy for a quarter of the resulting agent-games, 92% of it unconditional cooperation or unconditional defection. Tit-for-tat and win-stay lose-shift, the rules that carry the theoretical interest of the game, are named fifteen times more often than they are demonstrated, and where they are named the play

departs from them on more than two rounds in ten. The distribution over strategies is in any case not a property of the agent: it moves beyond a permutation null under every manipulation we apply, including a rescaling of the payoffs that cannot change the game. Mining the quarter the read-out will not name recovers a structure the canonical vocabulary has no room for, a two-phase rule switching once at a median of round two and running towards cooperation, and behind it a majority of play that is drift rather than any rule at all.

The practical recommendation follows from the first result alone and costs nothing to adopt. Run the deductive stage before the classifier, let the read-out say “no canonical rule explains this”, and publish the three shares and the departure count alongside any strategy label. Until that is routine, the field risks building an evolutionary game theory of artificial agents on names that were assigned rather than earned.

Data accessibility

The full corpus of 2,400 dyads, the analysis pipeline that produces every figure and table, the trained read-out checkpoint and the supplementary figure suite are available at [REPOSITORY URL]. Every number appearing in a figure is also written to a machine-readable CSV file in the same repository. A permanent archive with a DOI will be deposited on acceptance.

Competing interests

The authors declare no competing interests.

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